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Alzheimer’s Disease Stage Classification Using Brain MRI Images

IT 461: Practical Machine Learning | 2nd Semester 1447 H

Introduction

Alzheimer's disease affects memory and brain function, making early diagnosis very important for improving treatment and patient care.

MRI scans contain important structural brain information, but manual analysis can be difficult and time-consuming. In this project, deep learning models were used to automatically classify brain MRI images into Healthy or Diseased categories.

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Objective :

1. Build deep learning models for MRI classification.

2. Compare different CNN architectures.

3. Improve performance using transfer learning

4. Classify MRI images into:

- Healthy
- Diseased

Related Works

Study	Method	Dataset	Result
Thevar et al. (2025)	CNN, SVM, RF	OASIS	CNN preformed best
Hossain et al. (2024	VGG16, DenseNet121	ADNI	VGG16 achieved 95%
Mohammed et al. (2021)	AlexNet + SVM	OASIS	94%accuracy

These studies demonstrate that CNNs and transfer learning are highly effective for Alzheimer's MRI classification.

Dataset Overview

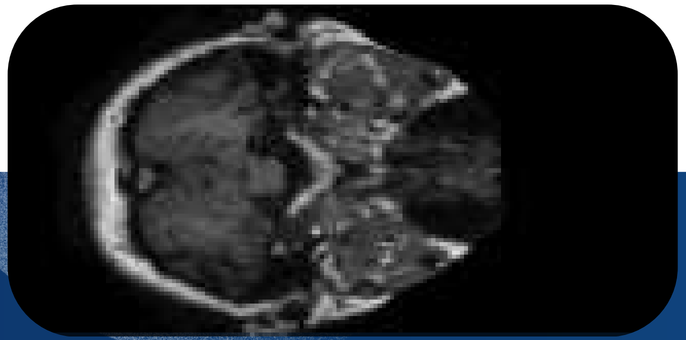
Characteristic	Value
Total Images	10,442 MRI scans
Unique Patients	68
Classes	4 (merged to 2 for binary)
Image Format	JPEG (grayscale)
Source	Kaggle

Class Distribution (after binarization):

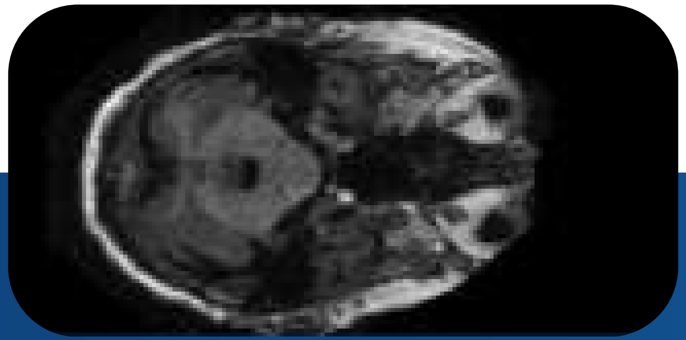
Healthy: ~3,000 images
Diseased: ~7,400

~10,400 MRI images from 68 patients

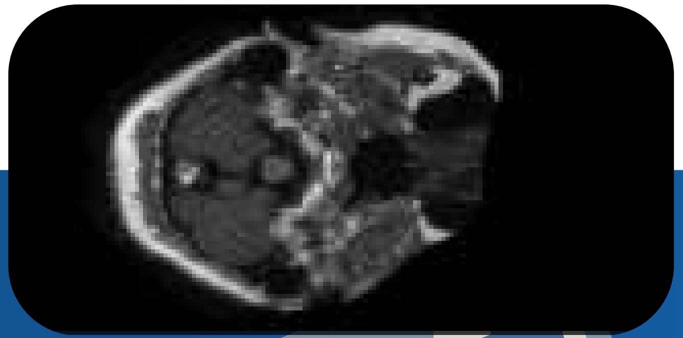
Non Demented



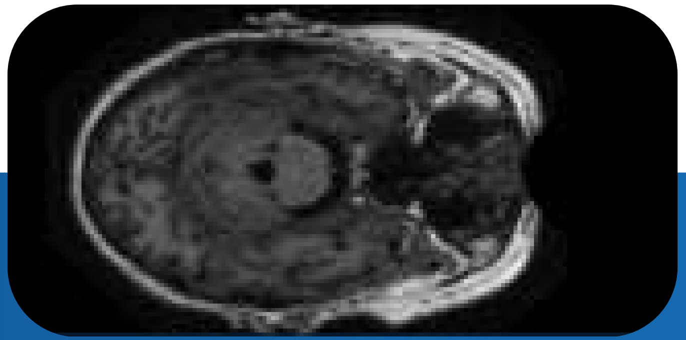
Mildly demented



Very Mild Demented



Moderate Demented



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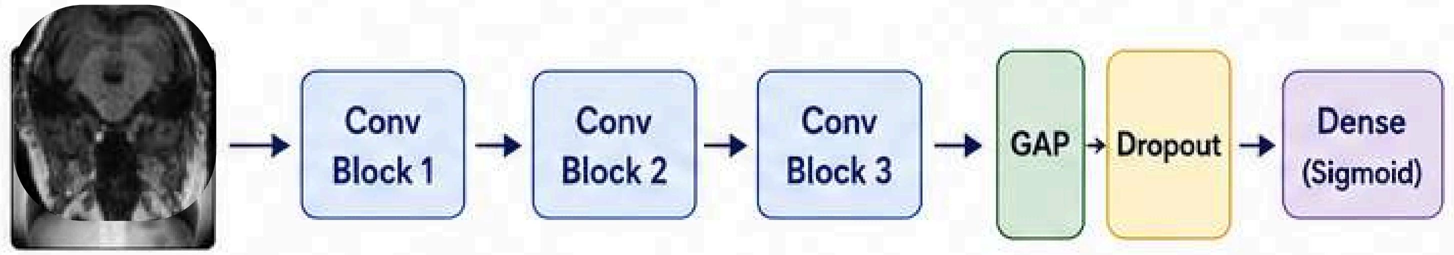
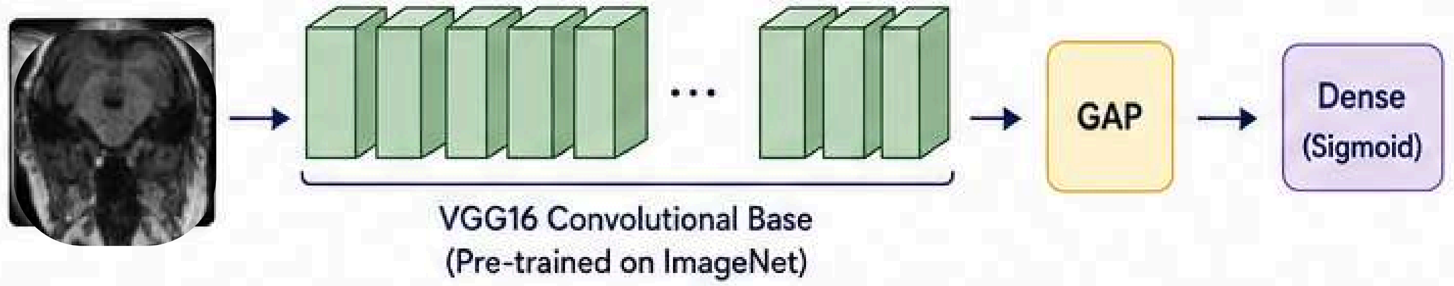
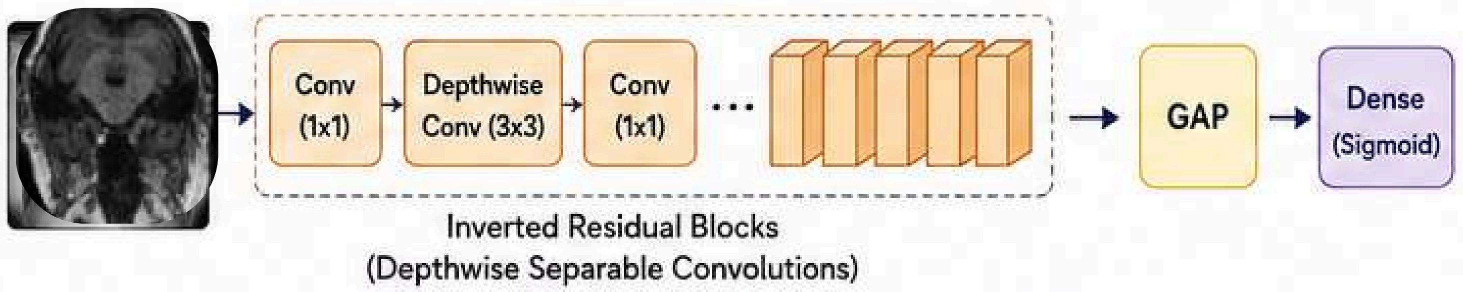
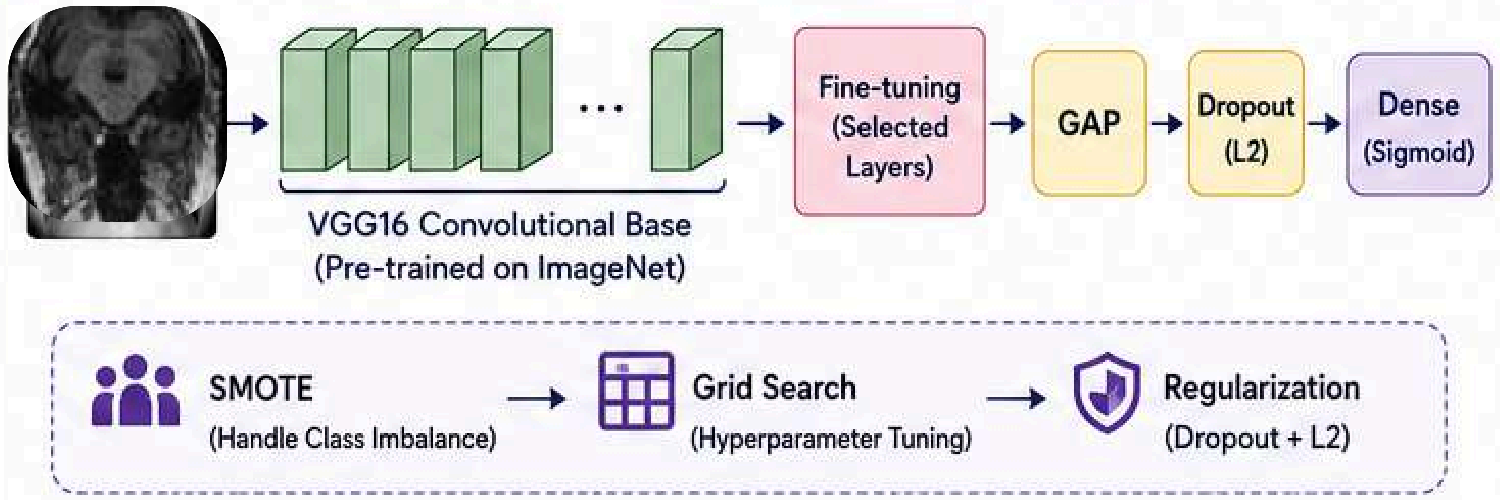
Methods : Model Architecture

Model	Type	Key Characteristics	Architecture / Overview
Custom CNN	Baseline (from scratch)	• 3 conv blocks • BatchNorm • Dropout	
VGG16	Transfer Learning	• 16 layers • Pre-trained on ImageNet	
MobileNetV2	Lightweight CNN	• Depthwise separable convolutions	
Enhanced VGG16	Transfer Learning + Tuning	• SMOTE • Grid Search • Regularization	

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Experimental setup

Training Configuration

Optimizer: Adam

Loss: Binary Cross-Entropy (Focal Loss for Custom CNN)

Batch Size: 32

Epochs: 15-20 (with early stopping)

Cross-Validation

Method:
5-fold patient-wise stratified

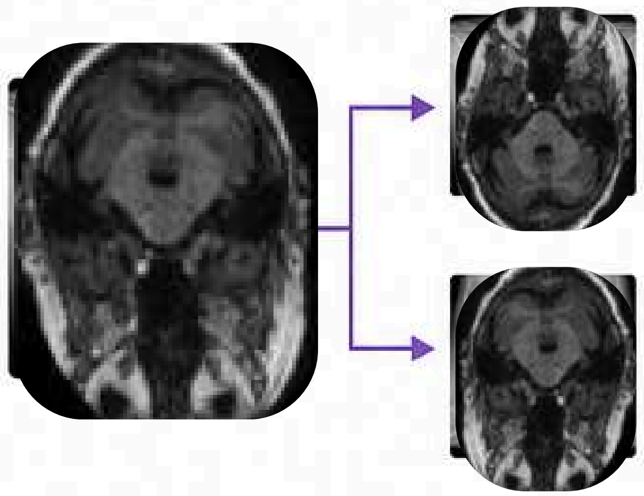
Why:
Prevents data leakage

	Split 1	Split 2	Split 3	Split 4	Split 5
Fold 1	Test	Train	Train	Train	Train
Fold 2	Train	Test	Train	Train	Train
Fold 3	Train	Train	Test	Train	Train
Fold 4	Train	Train	Train	Test	Train
Fold 5	Train	Train	Train	Train	Test

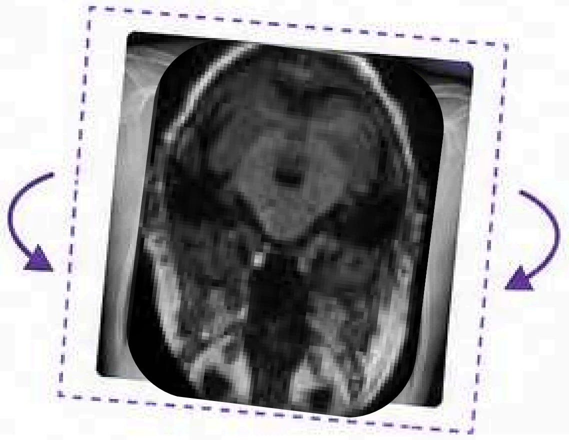
Train (80%) Test (20%)

Data Augmentation

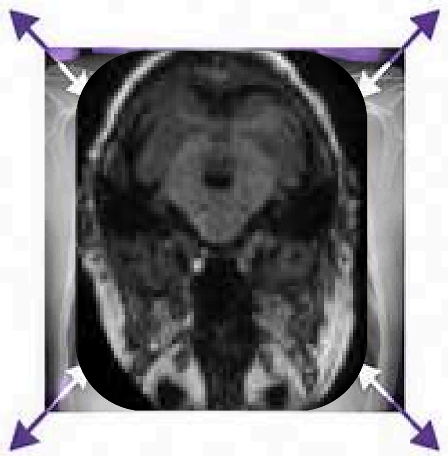
Random flips
(horizontal/vertical)



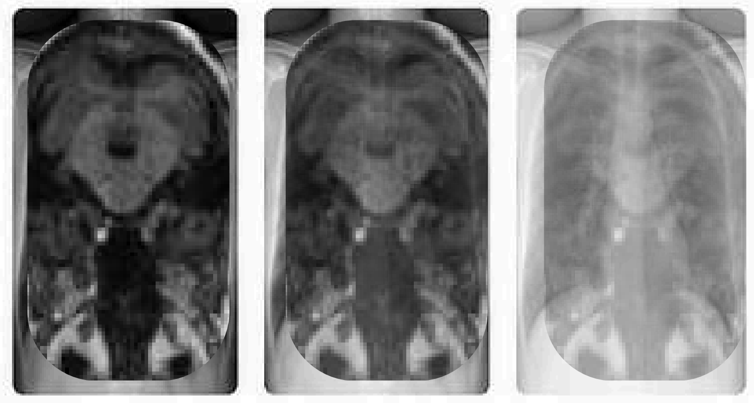
Rotation
(±10%)



Zoom
(±10%)



Contrast adjustment



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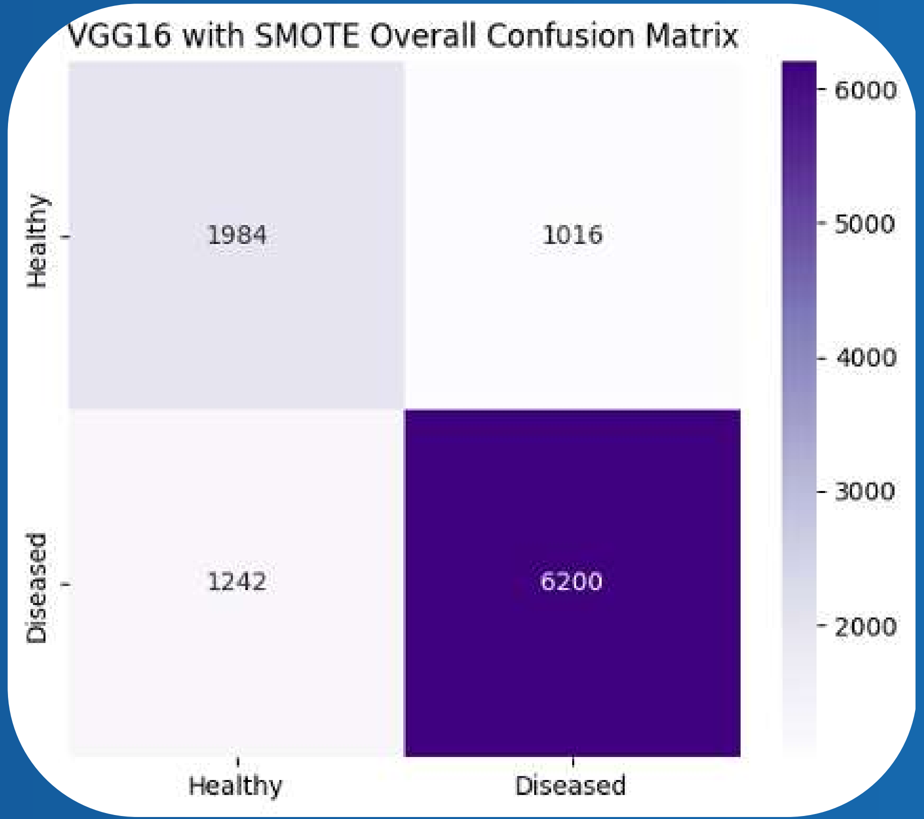
Results – Overall Performances

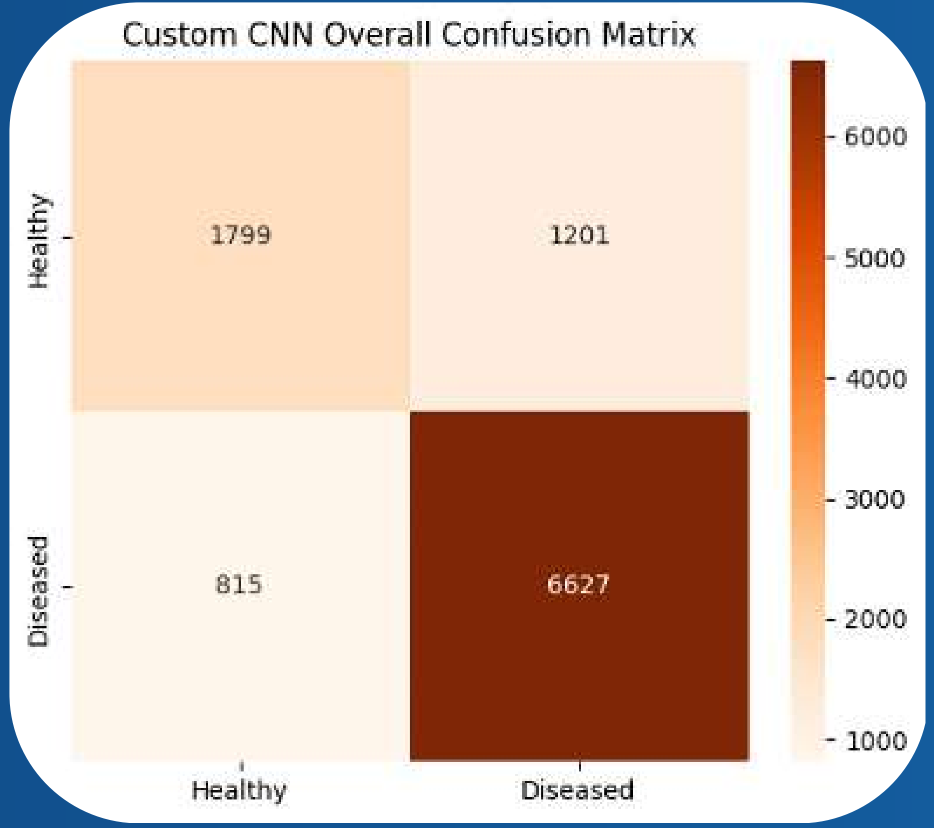
MODEL	ACCURACY	PRECISION (H / D)	RECALL (H / D)	F1-SCORE (H / D)
CUSTOM CNN	0.80	0.69 / 0.85	0.60 / 0.89	0.64 / 0.87
MOBILENETV2	0.60	0.37 / 0.78	0.55 / 0.63	0.44 / 0.69
STANDARD VGG16	0.75	0.58 / 0.83	0.56 / 0.83	0.57 / 0.83
ENHANCED VGG16 (SMOTE + GRID SEARCH)	0.78	0.62 / 0.86	0.66 / 0.83	0.64 / 0.85

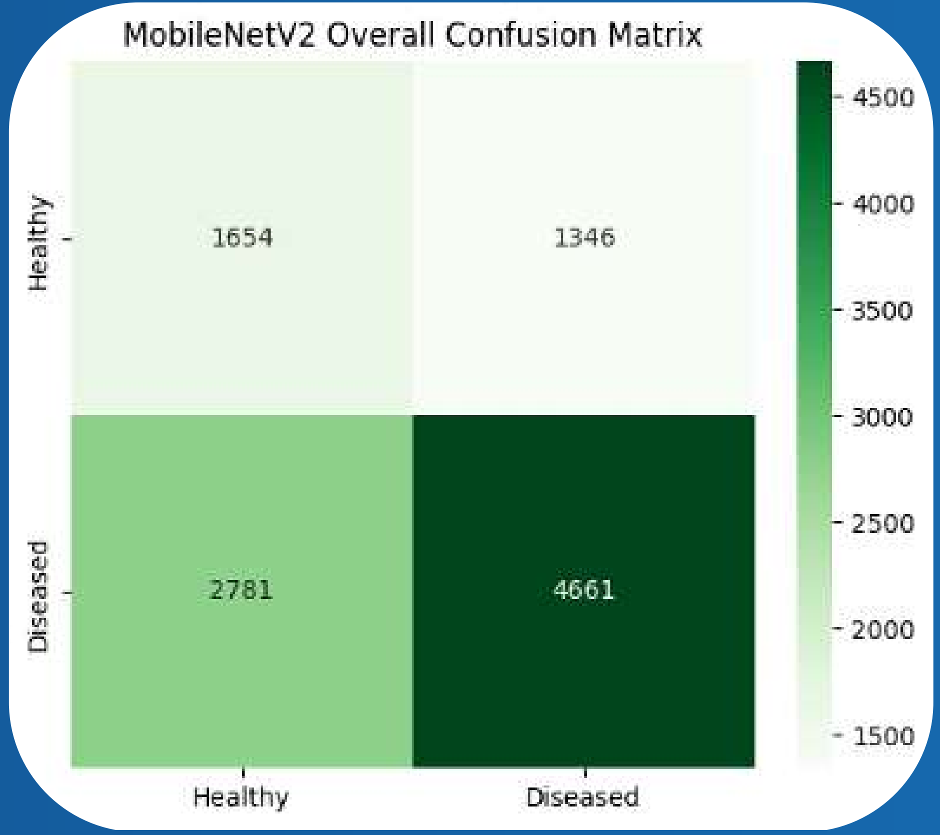
Key Finding: the Enhanced VGG16 achieved the best performance

Results – confusion matrixes

1. **Best Accuracy: Custom CNN achieved the highest accuracy (80%) with strong Alzheimer's detection (0.87 F1-score).**
2. **Highest Sensitivity: Custom CNN reached the highest diseased recall (0.89), detecting Alzheimer's cases effectively.**
3. **Most Balanced Model: Enhanced VGG16 provided the most reliable and balanced performance (78% accuracy) after SMOTE and tuning.**
4. **Optimization Benefit: Enhanced VGG16 outperformed Standard VGG16 (78% vs. 75%), showing the effectiveness of optimization techniques.**
5. **Weakest Model: MobileNetV2 showed the lowest performance (60% accuracy) due to underfitting.**







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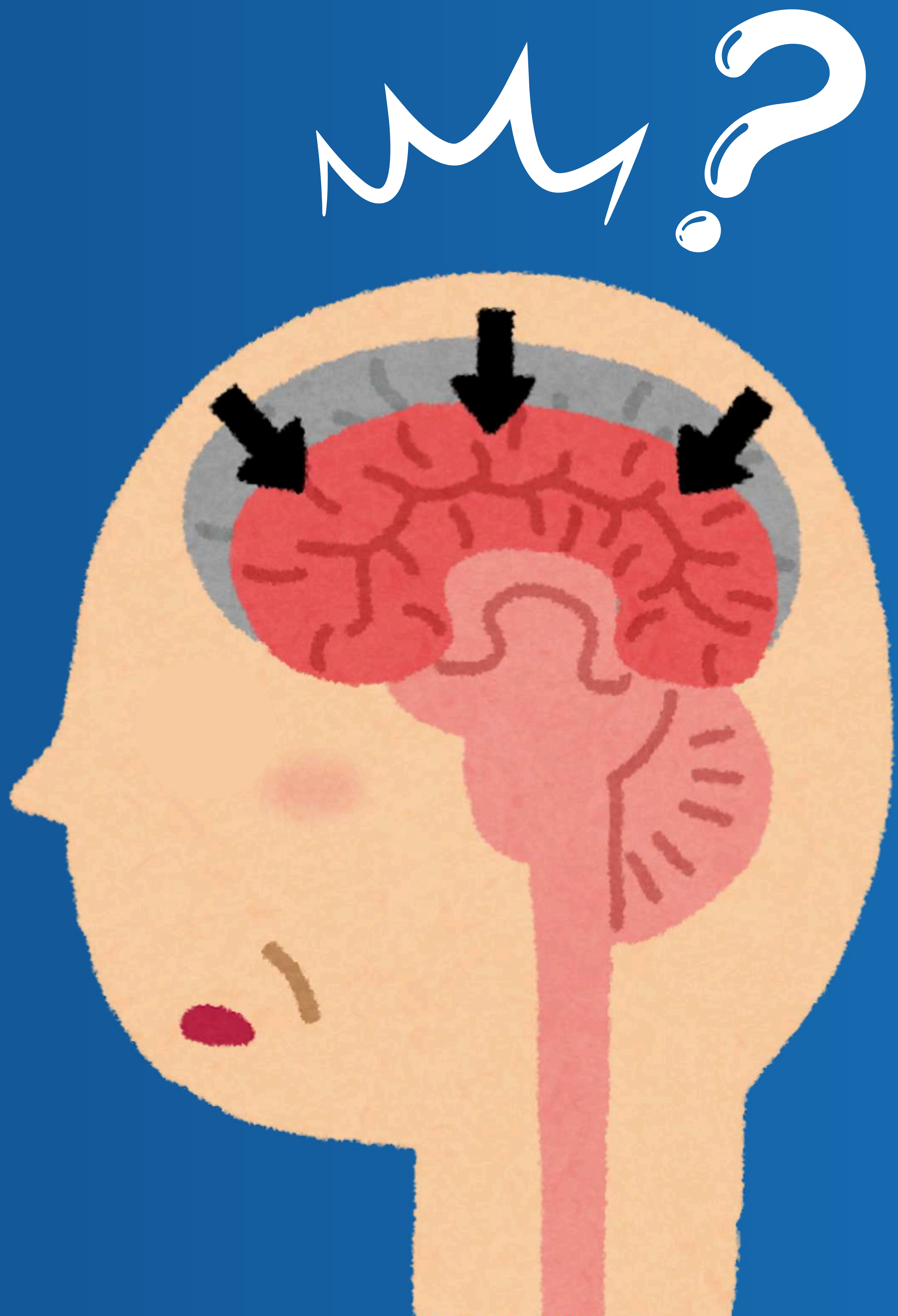
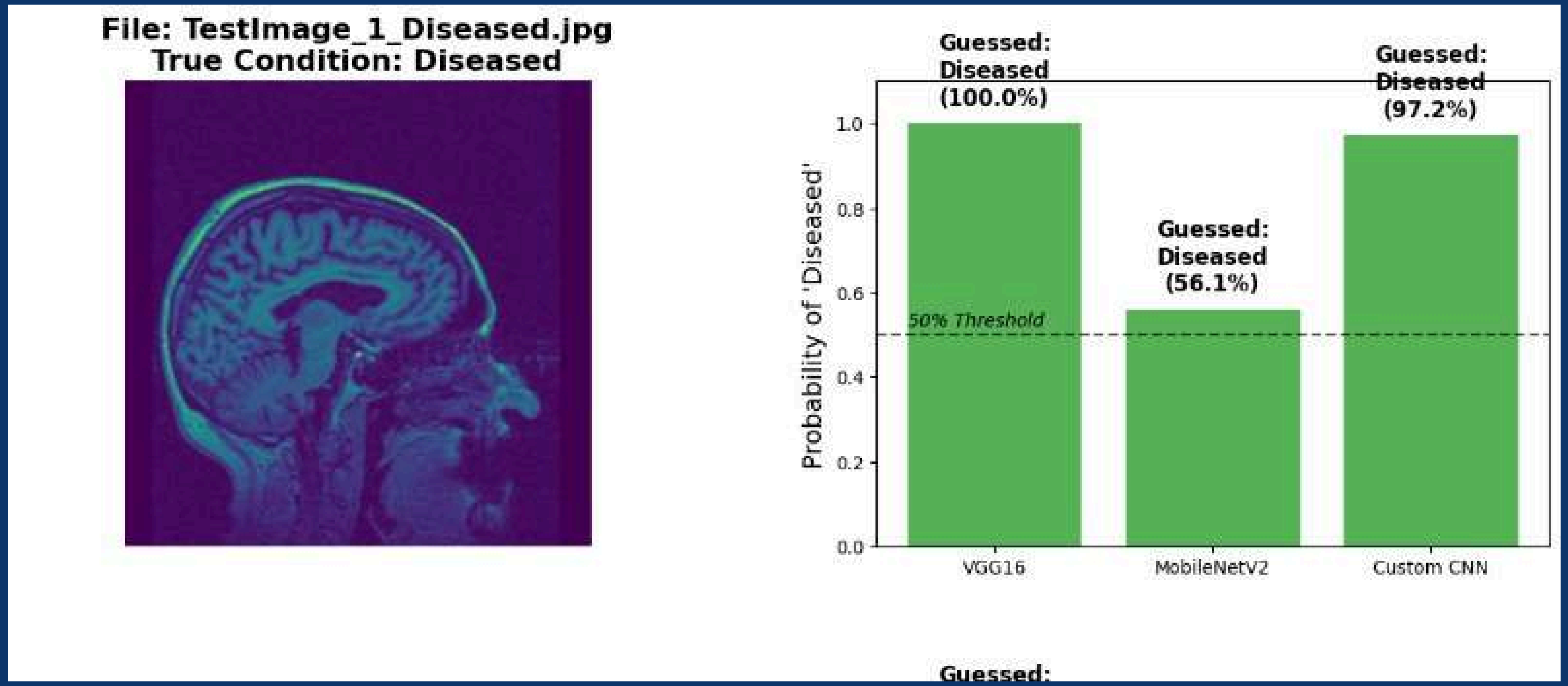
Conclusion

- Clinical Viability: Deep learning effectively classified Alzheimer’s MRI images with strong diseased-case detection.
- Optimization Impact: SMOTE, regularization, and hyperparameter tuning improved model balance and robustness.
- Top Performance doesn’t mean most reliab: Custom CNN:80% overall accuracy 89% diseased recall (highest sensitivity)

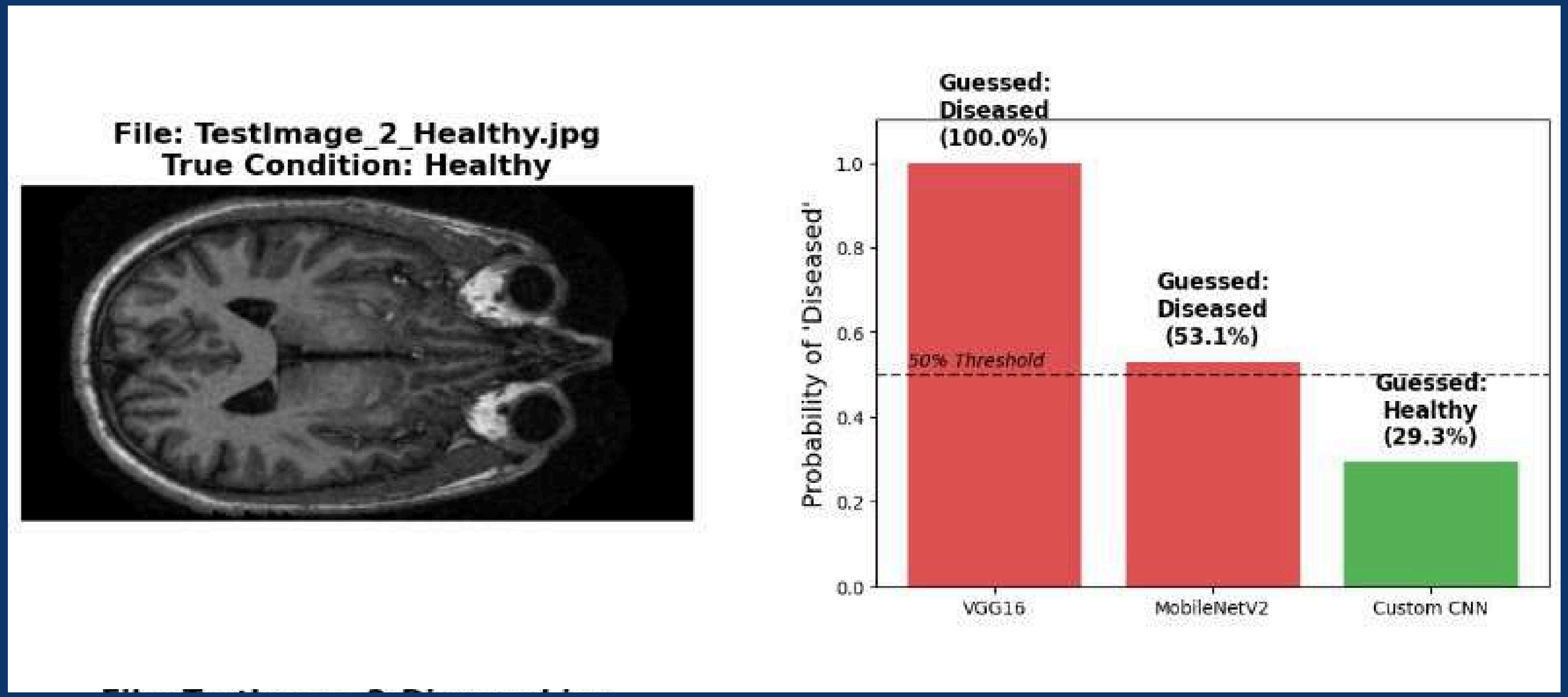
  Enhanced VGG16 achieved the most balanced
- Reliable Evaluation: Patient-wise cross-validation improved evaluation reliability and generalization.



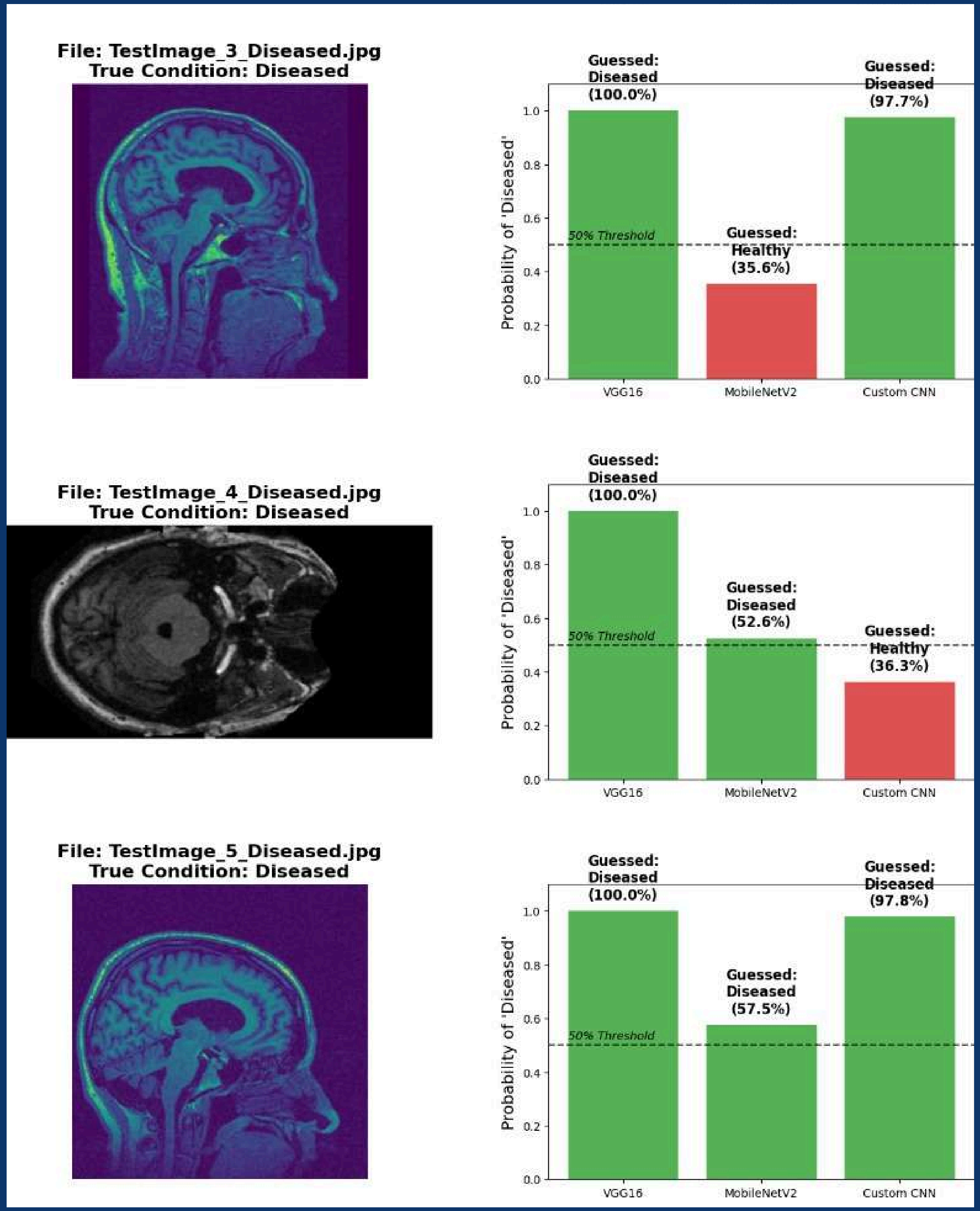
Live Demo :



File: TestImage_1_Diseased.jpg
True Condition: Diseased
Guessed: Diseased (100.0%)
Guessed: Diseased (56.1%)
Guessed: Diseased (97.2%)
50% Threshold
Probability of 'Diseased'
1.0
0.8
0.6
0.4
0.2
0.0
VGG16
MobileNetV2
Custom CNN
Guessed:



File: TestImage_2_Healthy.jpg
True Condition: Healthy
Guessed: Diseased (100.0%)
Guessed: Diseased (53.1%)
Guessed: Healthy (29.3%)
50% Threshold
Probability of 'Diseased'
1.0
0.8
0.6
0.4
0.2
0.0
VGG16
MobileNetV2
Custom CNN



File: TestImage_3_Diseased.jpg
True Condition: Diseased
Guessed: Diseased (100.0%)
Guessed: Healthy (35.6%)
Guessed: Diseased (97.7%)
50% Threshold
Probability of 'Diseased'
VGG16
MobileNetV2
Custom CNN
File: TestImage_4_Diseased.jpg
True Condition: Diseased
Guessed: Diseased (100.0%)
Guessed: Diseased (52.6%)
Guessed: Healthy (36.3%)
50% Threshold
Probability of 'Diseased'
VGG16
MobileNetV2
Custom CNN
File: TestImage_5_Diseased.jpg
True Condition: Diseased
Guessed: Diseased (100.0%)
Guessed: Diseased (57.5%)
Guessed: Diseased (97.8%)
50% Threshold
Probability of 'Diseased'
VGG16
MobileNetV2
Custom CNN

Thank You